z5418531 RCE due to EBMD

by Nathan Do

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z5418531

Recurrent Corneal Erosion due to Epithelial Basement Membrane Dystrophy

Case Report

Word count: 1794 (excluding abstract, tables and references)

Abstract:
Recurrent Corneal Erosion (hereafter: RCE) is a chronic, relapsing disorder that is characterised by a dysfunctional epithelial ecosystem, wherein the corneal epithelium fails to adhere to the underlying basement membrane and Bowman's layer **(Lin et al., 2019; Das & Seitz, 2008)**. The major etiologies of RCE are separated into intrinsic (primary) or acquired (secondary) causes. Secondary causes of RCE are acquired later in life due to injury, disease or inflammation; these include: mechanical trauma, corneal degenerations, ocular infections, and ocular surface and lid disease **(Ramamurthi et al., 2005).** Primary reasons for RCE are genetically mediated and can develop in multiple corneal locations; these are: epithelial dystrophies, bowman layer dystrophies, stromal dystrophies, and endothelial dystrophies. Epithelial Basement Membrane Dystrophy (hereafter: EBMD) is the most common hereditary anterior corneal dystrophy and accounts for about 19-29% of RCE cases **(Laibson, 2010).**

EBMD is also known as Cogan's microcystic or map-dot-fingerprint dystrophy, and is the result of the abnormal development of the corneal basement membrane. Trapped cells and redundant tissue layers lead to hallmark signs of 'maps', 'dots' and 'fingerprints' on the cornea, which are best seen via broad-angled illumination and retroillumination **(Laibson, 2010; Das, 2025)**. People with EBMD often report symptoms of blurred vision, photophobia, and a foreign body sensation **(Xu et al., 2019; Brown & Bron, 1976).**

The treatment of RCE involves two stages: promoting re-epithelisation and preventing recurrence through unpacking the underlying cause. The treatment of an acute RCE involves an antibiotic prophylaxis to prevent serious complications like microbial keratitis, lubricants and can introduce pain management strategies **(Xu et al., 2012)**. Considering that the underlying cause of the RCE is EBMD, there should also be a regard for long term management strategies to optimise prognosis. This report describes the presentation, diagnosis, and management of a RCE that is suspected to be due to EBMD in a 65-year-old female.

Case History:
LZ is a 65-year-old East Asian female who presented to the clinic with an acute onset pain in her right eye that was most severe upon waking that same morning. She reported an associated foreign body sensation, but denied any change in vision, headache, flashes, or associated discharge. LZ described a longstanding history of mild dry eye symptoms bilaterally and she states that she experiences relief with lubricant drops. She has a longstanding history of floaters that have remained stable in frequency and density. She denied any recent illness, ocular trauma, foreign body, or contact lens wear, and had no personal or family history of similar events. Her ocular history was significant for EBMD – which she claims to have existed since childhood – and she reports that her father has an unspecified corneal dystrophy. Her medical history included hypertension which she manages with telmisartan (Micardis). She is also taking Ibuprofen to manage her ocular pain.

LZ's last eye examination was in 2025. Here, she achieved the same visual acuities as in her initial and follow-up appointment (6/12 OD, 6/20 OS).

Clinical Findings (Day 1)	
Chief Complaint	CC: recent onset painful eye upon waking (right eye). Since this morning and does not recall having similar episodes before. Pain is described as a sharp stabbing pain but has subsided with Ibuprofen. Still feels uncomfortable.

History and Sx	Current Rx: 2024 SVD + SVN. Not a contact lens wearer. Nil changes in vision. D OK N OK. (-) HA, diplopia, asthenopia (-) discharge, recent illness (+) red and itchy eyes; sometimes 9 (+) dry eyes; finds relief with lubricant drops, but has not used as often (-) flashes (+) longstanding floaters, have not changed
POcHx	(+) EBMD; diagnosed in 2023 at SEH, but suspects she has had this since childhood (-) AMD (-) glaucoma (-) eye inj, surgery
FHx	(-) AMD (-) glaucoma Father - (+) corneal dystrophy (type unspecified)
PMHx	Medications: Micardis, Ibuprofen 400mg (+) High BP, (-) diabetes (-) allergies
Entering Visual Acuities (aided)	-5.50/-2.25x67 (VA: 6/12) 10 -9.25/-1.00x155 (VA: 6/20) Nil change in BCVAs.
IOP (NCT at 11:40 AM)	OD: 9/10/9 mmHg, OS: 9/10/9 mmHg
Pupils	PERRL (-) RAPD

Anterior Eye Biomicroscopy	OD	OS
Lids and lashes	Mild meibomian gland capping I>S	Mild meibomian gland capping I>S
Iris	Clear	Clear
Conjuctiva	G2 redness temporally and nasally	G1-2 redness temporally and nasally
Cornea	- Small, grey intraepithelial opacities. - Geographic map lines. Fluorescein Assessment - 3 x 2 mm inferior-paracentral epithelial defect. Negative staining along edges. - Scattered micropunctate staining. - Negative staining on EBMD lesions.	- Small, grey intraepithelial opacities. - Geographic map lines. Fluorescein Assessment - Scattered micropunctate staining. - Negative staining on EBMD lesions.
Anterior Chamber	Deep and quiet	Deep and quiet
Lens	G2 NS (LOCS III)	G2 NS (LOCS III)

Clinical Findings (Day 2)	
Chief Complaint	CC: 1 day follow up appointment. Reported relief with sx. Nil pain. Nil discharge. Nil changes in vision.

History and Sx	Current Rx: 2024 SVD + SVN. Not a contact lens wearer. Nil changes in vision. D OK N OK. (-) HA, diplopia, asthenopia (-) discharge, recent illness (+) red and itchy eyes; sometimes (+) dry eyes; but finds relief with lubricant drops, but has not been used as often (-) flashes (+) longstanding floaters, have not changed
POcHx	(+) EBMD; diagnosed in 2023 at SEH, but suspects she has had this since childhood (-) AMD (-) glaucoma (-) eye inj, surgery
FHx	Father - (+) corneal dystrophy (type unspecified) (-) AMD (-) glaucoma
PMHx	Medications: Micardis, Ibuprofen 400mg (+) High BP, (-) diabetes (-) allergies
Entering Visual Acuities (Aided)	-5.50/-2.25x67 (VA: 6/12) -9.25/-1.00x155 (VA: 6/20)
IOP (NCT at 10:01 AM)	OD: 11/10/11 mmHg, OS: 12/11/10 mmHg
Pupils	PERRL (-) RAPD

Anterior Eye Biomicroscopy	OD	OS
Lids and lashes	Mild meibomian gland capping I > S	Mild meibomian gland capping I > S
Iris	Clear	Clear
Conjuctiva	G1 redness T + N (improved from grade 2)	G1 redness T + N (improved from grade 2)
Cornea	- Small, grey intraepithelial opacities. - Geographic map lines. Fluorescein Assessment - 1 x 1 mm epithelial defect. Negative staining around the defect. - Scattered micropunctate staining. - Negative staining on EBMD lesions.	- Small, grey intraepithelial opacities. - Geographic map lines. Fluorescein Assessment - Scattered micropunctate staining. - Negative staining on EBMD lesions.
Anterior Chamber	Deep and quiet	Deep and quiet
Lens	G2 NS (LOCS III)	G2 NS (LOCS III)

Note: Further follow-up appointments were conducted outside the private practice placement hours.

Differentials:

Differential	Reasons For	Reasons Against

Meesmann Corneal Dystrophy **(Lin et al., 2019; Rosenberg et al., 2000)**	Recurrent erosion-type presentation with bilateral corneal changes. LZ claims that her corneal dystrophy was from childhood – a timeframe in which Meesmann tends to develop.	Meesmann dystrophy typically presents with intraepithelial microcysts that extend to the limbus. It lacks the geographic map lines seen in this patient. EBMD is frequently asymptomatic until the 3rd to 5th decade of life, which aligns with LZ's clinical profile, despite claims of childhood onset.
Dry Eye Disease / Filamentary Keratitis **(Miller et al., 2019)**	Longstanding history of dry eye and lubricant use. MGD was also noted. Scattered micropunctate staining presenting in both eyes.	Negative staining along epithelial defect edges reflects a structurally abnormal epithelial-basement membrane adhesion. There were no mucous filaments observed. Dry eye is most likely a cofactor rather than a primary cause.
Microbial Keratitis **(Das, 2025)**	Follows characteristic signs and symptoms such as pain, foreign body sensation, and corneal epithelial defect.	Upon slit lamp biomicroscopy, there is the absence of a stromal infiltrate or corneal opacity. The anterior chamber showed no reaction. Furthermore, the patient reported no discharge.
Herpes Simplex Virus Epithelial Keratitis **(Ramamurthi et al., 2005)**	Typically presents unilaterally at first. Aligns with patient presentation of painful red eye with epithelial irregularity.	Absence of dendritic or geographic ulcer. There was no reduced corneal sensation or pseudodendrites on fluorescein staining.
Traumatic (isolated) Corneal Erosion **(Xu et al., 2012)**	Trauma is a common cause of RCE. Pain upon awakening is a classic presenting feature.	No history of recent trauma, foreign body, or contact lens wear. The presence of bilateral map-dot-fingerprint changes strongly favours underlying dystrophy aetiology.
Contact Lens-Related Issues **(Miller et al., 2019).**	Peripheral and focal epithelial defects can mimic RCE. Contact lenses are a risk factor for RCE.	LZ reports no contact lens wear.

Diagnosis:
RCE due to Epithelial Basement Membrane Dystrophy

The diagnosis of RCE aligns with the classic history of severe, ocular pain that occurs immediately after waking or eye opening. This is because the force of the moving e[illegible] greater than the weak attachment of the epithelium, resulting in the sensation of a sharp pain **(Miller et al., 2019).** The erosion is confirmed by clinical findings of an epithelial defect in the right eye that had positive fluorescein staining inferior and paracentrally in both the initial and follow up visit **(Brown & Bron, 1976)**. In LZ's case, the presence of bilateral map-dot-fingerprint changes under slit lamp examination aligns with her pre-established 2023 diagnosis of EBMD. These EBMD lesions also demonstrated negative staining on fluorescein assessment **(Laibson, 2010).**

EBMD is the most likely cause of [illegible] RCE due to structural defects and biological enzymes. The epithelium is tightly nailed down by anchor proteins called hemidesmosomes, which are formed incorrectly in EBMD, making the eyelid prone to shearing off the loose surface and causing RCEs **(Miller et al., 2019).** Moreover, because of the surface instability, high levels of enzymes called

Matrix Metalloproteinases (MMPs) are produced, which become overactive and can further dissolve anchoring proteins **(Jadczyk-Sorek et al., 2023; Ramamurthi et al., 2005).** As a result, this creates a cycle that leads to extensive tissue breakdown.

Regarding pathophysiology, around 10% of those with EBMD develop a symptomatic erosion **(Miller et al., 2019)**. The condition has an estimated prevalence between 5% and 18% **(Kim & Chung, 2024)**, is more prevalent in females and typically becomes symptomatic in the third to fifth decade – though the underlying dot and fingerprint changes may be present earlier **(Laibson, 2010)**. While EBMD typically presents sporadically, there is evidence that an inherited TGFBI gene may cause the condition **(Boutboul et al., 2006)**, which accounts for LZ's reported childhood-onset disease and her father's corneal dystrophy.

Management:
In managing RCE as a result of EBMD, the aims are to promote epithelial healing, reduce ocular surface friction and inflammation, and address the underlying basement membrane pathology to reduce the likelihood of future recurrence. The initial conservative approach is supported by literature, beginning with lubrication and prophylactic antibiotic cover, and progressing to pharmacological interventions. Given the severity, surgical interventions may also be considered.

1. For the acute erosion LZ was prescribed **Chloramphenicol 0.5% 10mL eye drops**, 1 drop in the right eye four times a day for 5-7 days (or until the defect fully closes) **(Hykin et al., 1994)**. Pharmacologically, chloramphenicol is a broad-spectrum bacteriostatic antibiotic that inhibits bacterial protein synthesis by binding to the 50S ribosomal subunit **(Oong & Tadi, 2021).** This was prescribed as a short-term prophylaxis against microbial keratitis, but may create side effects of transient stinging. A major contraindication is a personal or family history of blood dyscrasias. LZ was a suitable candidate given her history of no known allergies or blood disorders.
2. Aggressive lubrication was also recommended. **Hyloforte 10mL (0.2% sodium hyaluronate)** preservative-free lubricating drops were prescribed, with 1 drop into both eyes every 2-4 hours. Hyloforte was suggested because of its active ingredient, sodium hyaluronate, which actively promotes cellular migration and re-epithelisation without the toxic effects of preservatives on a compromised cornea **(Gomes et al., 2004; Tse et al., 2020).**
3. For overnight protection, **Vita-POS 5mg lubricating ointment (retinol palmitate)** was recommended, with 1cm to be applied into the lower lid of both eyes at bedtime. This was applied to prevent the eyelids from adhering to and shearing the loose epithelium upon awakening **(Hykin et al., 1994).**
4. LZ was counselled on the use of an oral analgesic for pain management. Since she was already taking this medication to manage her symptoms, we advised LZ to continue following medication guidelines. More specifically, that is: taking **Ibuprofen 400mg** orally every 4-6 hours as needed for the pain, but not exceeding the day limit of 1200 mg **(Miller et al., 2019)**. Ibuprofen is an NSAID that inhibits cyclooxygenase enzymes, reducing prostaglandin synthesis, and therefore, inflammation. This medication is contraindicated in patients with active gastric ulcers or severe renal impairment, which LZ does not report having.

Since LZ was managing the pain well with an oral analgesic, no further therapy was prescribed for her pain. Given severe pain, a further strategy that can be used to manage pain is the in-office 1 drop of topical cycloplegic into her right eye – typically cyclopentolate 1%. After instilling the drop, it is important the clinician blocks systemic absorption via punctual occlusion.The muscarinic receptor antagonist paralyses the ciliary muscle and stops the reflex ciliary spasm caused by exposed

corneal nerves **(Miller et al., 2019)**. Side effects include photophobia, stinging upon instillation, and blurred near vision. It is contraindicated in populations of down syndrome and brain damage due to risk of central nervous system toxicity and it is avoided in pregnancy and breastfeeding **(Contreras-Salinas et al., 2022).**

For management of MGD, LZ was advised to use a 'Bruder Moist Heat Eye Compress' for 10 minutes every day and lid massage to stimulate meibomian gland expression and relieve dry eye symptoms.

A strict follow-up protocol was initiated, with the first review at 1 day to ensure the epithelial defect has reduced in size and to rule out secondary microbial keratitis. Subsequent reviews were scheduled every 1-2 days until complete re-epithelisation was noted. In terms of the duration, after re-epithelisation, she may discontinue antibiotics and pain management, however, she should continue lubrication and ointment to deter recurrence.

Discussion/Prognosis:
RCE is a chronic condition, with episodes recurring intermittently over months or years, especially if the underlying EBMD is not addressed. While the acute epithelial defect is expected to heal within several days with the current conservative treatment, the presence of EBMD means that hemidesmosomal adhesion remains compromised, and further erosions are very likely **(Laibson, 2010).**

In acute cases, the primary short-term goal is preventing microbial keratitis. Prophylactic antibiotic use demonstrates a great prognosis for preventing infection while the epithelium heals. A randomised control trial of 82 acute RCE participants utilised a regimen that involved topical chloramphenicol ointment – this led to zero instances of infection and 95% of the RCEs resolving within 5 days **(Hykin et al., 1994).** LZ showed improvements in just one day as the epithelial defect reduced in size and she no longer experienced the accompanying pain.

To prevent recurrences in the long term, the compromised basement membrane should be addressed. This begins with the [17] ual use of overnight and daytime lubrication to allow the hemidesmosomes to regenerate. The lubricating regimen proves to have the [18] effectiveness of hypertonic saline ointment, which is another common management for RCE **(Watson & Leung, 2018).**

If conventional lubrication fails, advanced conservative interventions are as follows:

1. LZ can initiate **autologous serum drops** 6 times a day to both eyes, which provide critical adhesion proteins and growth factors that directly contribute to healing the corneal epithelium **(Ramamurthi et al., 2005).** In a prospective study of 33 participants that had failed conventional therapy, 85% achieved complete epithelial healing with no relapse over a 30-month follow up period **(Lin et al., 2019).**
2. **Ikervis (Ciclosporin 0.1%)** drops are an immunosuppressant that inhibits T-cell activation in order to reduce ocular surface inflammation. These would be used once daily to both eyes at bedtime, and may cause site irritation and eyelid erythema **(Labetoulle et al., 2022)**. This is contraindicated if LZ's erosion becomes infected, due to the immunosuppression. In a retrospective case review of dry eye disease and RCE patients that were unresponsive to other treatments, all patients demonstrated an absence of new RCE episodes and improved tear film stability **(Napoli et al., 2019).**
3. A **Bandage Contact Lens (BCL)** is a silicone hydrogel BCL that acts as a physical barrier to prevent the eyelid from adhering to the healing epithelium. If a BCL is used, prophylactic

topical antibiotics (ciprofloxacin 0.3% QID) is prescribed, and the patient will be monitored closely because of the elevated risk of infection and microbial keratitis. A minimum of 6 weeks required to see a remodelling of the epithelial basement membrane **(Das, 2025).** Scleral lenses may also be considered if the BCL fails, as they vault the cornea entirely and provide a constant liquid reservoir **(Xu et al., 2012)**.

4. Finally, further MGD treatments can be considered. This involves **oral doxycycline 50mg** twice daily alongside **topical fluorometholone (FML) 0.1%** three times a day for 8 weeks. Doxycycline is a tetracycline antibiotic that inhibits MMP-9 activity – reducing basement membrane degradation **(Brown et al., 1976)**. Side effects of doxycycline include an upset gastrointestinal tract and photosensitivity. It is contraindicated in pregnancy. **(Patel & Parmar, 2023).** FML carries the risk of intraocular pressure spikes and cataract formation, and therefore requires pressure mentoring. It is contraindicated in active ocular infection **(Cornell, 1987)**.

When conservative management fails, definitive treatment is surgical.

Surgery	Information
Epithelial debridement	Involves manually removing loose epithelium and yields a 30-50% success rate **(Miller et al., 2019).**
Anterior Stromal Puncture (ASP)	Procedure that uses a needle to create microscopic scars to anchor the epithelium. Success rates of this technique are generally 80% **(Miller et al., 2019)**, however, can cause subepithelial scarring which results in glare and reduced visual acuity **(Das, 2025)**.
Phototherapeutic keratotectomy (PTK)	An excimer laser ablation of the superficial stroma, and creates a stromal bed for hemidesmosome adhesion **(Cavanaugh et al., 1999)**. A systematic review and meta-analysis states this achieves a 80-90% success rate.
Diamond Burr Polishing	A highly effective alternative to PTK for removing the faulty basement membrane and creating a stable foundation for the corneal epithelium **(Xu et al, 2012; Das, 2025)**.

Overall, while the prognosis for the conservative management of RCEs is favourable in the acute phase, EBMD carries a high long term recurrence rate, and continued monitoring and escalation of therapy may be required if conservative management is insufficient.

Conclusion:

LZ was diagnosed with RCE in her right eye, secondary to EBMD, based on a classic history of pain on waking, a corneal epithelial defect, and bilateral map-dot-fingerprint changes on slit lamp examination. She was managed with prophylactic choramphenicol, intensive sodium hyaluronate lubrication every 2-4 hours, night-time vita-POS, and analgesic pain control. A follow-up schedule (1 day, then every 1-2 days until re-epithelisation) was established. As EBMD is a chronic condition, there is a high recurrence rate under conservative management. LZ was set clear expectations and counselled on further pharmacological options and surgical interventions if conservative treatment was non-responsive.

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z5418531 RCE due to EBMD

GRADEMARK REPORT

FINAL GRADE

80 /100

GENERAL COMMENTS

1. Abstract:

Quite well written but alos see comments.

2. Case History:

See comments. Was the patient previously examined 2025 in the same clinic?

3. Clinical Findings:

Please define acronyms before using them. did anyone prescribe the Ibuprofen? If so, who and when? It would have been helpful if you were able to include images of the presentation to confirm your findings. Why is VA more reduced in the LE rather than the RE (affected eye)? Np stromal glow with fluorescein in the RE? Much of the data presented for Day 2 appears to be a repeat of Day 1 findings, so perhaps just presenting the changes form Day 1 might have saved you some effort. What was the management of the condition which produced the Day 2 results (a brief summary might have been helpful).

4. Differentials:

Quite well written and considered. However, if ruling out HSV, you might have also considered Zoster.

5. Diagnosis:

See comments.

6. Management:

The use of chloramphenicol is interesting as the solution is preserved. Do you think that might affect the corneal healing rate? Of the possible topical antibiotics available why do you think chloramphenicol in particular was chosen? The same could be said for Vita-POS. Any thoughts re the longterm use of the Ibuprofen? Would you consider the use of a bandage contact lens for a case like this to provide immediate relief?

7. Discussion:

Is the the primary short term goal to prevent microbial keratitis or provide the patient with a means of relieving the acute pain? A good summary of surgical option with some optometrists apparently performing ASP in Sydney. Did you have any information re the way SEH had treated this condition previously?

8. Conclusion:

Concise. What do you think is the overall prognosis for this patient?

9. References:

A good list of relevant references.

Comment 1

typo?

Comment 2

no pain?

Comment 3

bandage soft contact lenses too?

Comment 4

which drops and how frequently were these drops used?

Comment 5

no ocular trauma at all during her life?

Comment 6

in one eye or both eyes?

Comment 7

from the EBMD or some other cause?

Comment 8

by whom?

PAGE 3

Comment 9

allergy? seasonal? Hay fever?

Comment 10

pinhole VAs?

PAGE 4

PAGE 5

Comment 11

not sure this quite makes sense. wouldn't the pain suggest stimulation of the corneal nerves?

Comment 12

because?

Comment 13

teresting anatomical term

PAGE 6

Comment 14

any thoughts as to what these might be?

Comment 15

would you consider the ointment form for this patient?

Comment 16

any information re pharmacology, actions etc?

PAGE 7

Comment 17

how does this occur?

Comment 18

but is this availbabe in Australia?

PAGE 8

PAGE 9

PAGE 10

PAGE 11

RUBRIC: 2018 OCULAR THERAPEUTICS 16 / 20

DIAGNOSIS (20%) 15 / 20

SCALE 1 (0)	Diagnosis does not relate to signs and symptoms.
SCALE 2 (5)	Key signs and symptoms included but not linked to the diagnosis.
SCALE 3 (10)	Diagnosis linked to signs and symptoms. Little information on pathophysiology or epidemilology.
SCALE 4 (15)	Signs and symptoms linked to diagnosis. Accurate description of either pathophysiology or epidemiology.
SCALE 5 (20)	Signs and symptoms linked to diagnosis. Pathophysiology and epidemiology accurately described.

DIFFERENTIAL (20%) 15 / 20

SCALE 1 (0)	No differential diagnosis included.
SCALE 2 (5)	Important differentials not considered for the specific case presentation.
SCALE 3 (10)	Important differentials included but were not all clearly excluded or managed for the specific case presentation
SCALE 4 (15)	Important differentials clearly excluded or managed for the specific case presentation.
SCALE 5 (20)	All differentials clearly excluded or managed for the specific case presentation.

MANAGEMENT (20%) 15 / 20

SCALE 1 (0)	Unsafe management or major areas of management plan missing. No prescription included.
SCALE 2 (5)	Important aspects of management plan missing. Medical interventions include insufficient information on instructions, formulation, dosage, duration, pharmacology, side effects and/or contraindications. Unclear how discussed management relates to presented case. Written prescription missing elements making it invalid.
SCALE 3 (10)	Basic management plan described. Important medical interventions included. Some information provided on instructions, formulation, dosage, duration, pharmacology, side effects and contraindications. Some application of discussed management to presented case. Written prescription includes major elements but may require clarification.
SCALE 4 (15)	Comprehensive management plan described. Medical interventions include the majority of information regarding instructions, formulation, dosage, duration, pharmacology, side effects and contraindications. Application of management options to specific case presented. Written prescription includes all elements necessary to be valid.
SCALE 5 (20)	Comprehensive management plan described including medical and non-medical options. All medical interventions include formulation, instructions, dosage, duration, pharmacology, side effects and contraindications. Specific discussion of management of presented case included. Critical analysis of management performed. Written prescription includes all elements to be valid.

PROGNOSIS (20%) 15 / 20

SCALE 1 (0)	No prognosis given
SCALE 2 (5)	Little information on likely treatment outcomes, natural history of disease or long term management
SCALE 3 (10)	Basic prognostic information included on either likely treatment outcomes, natural history, or impact of long term management.
SCALE 4 (15)	Good prognostic information. Some more information required on either likely treatment outcomes, natural history, or impact of long term management.
SCALE 5 (20)	All likely outcomes of treatment described. Appropriate long term management included. Complete natural history of the disease.

LITERATURE (20%) 20 / 20

SCALE 1 (0)	Management and prognosis mostly supported by text books or other material of a similar, or lower, standard. Significant errors in referencing in text and in reference list or no references at all.
SCALE 2 (5)	Management and prognosis mostly supported by research but higher quality evidence was available in many cases. Obvious errors in referencing in text or in reference list.
SCALE 3 (10)	Management and prognosis mostly supported by high quality evidence. Key concepts referenced. Several important publications on the topic were not acknowledged. Some errors or issues with referencing or reference list.
SCALE 4 (15)	Management and prognosis mostly supported by best available evidence. Key concepts referenced. However, a major publication or set of guidelines in the field was not acknowledged. Majority of referencing appropriate for academic submissions.
SCALE 5 (20)	Management and prognosis supported by the best available evidence. Key concepts referenced. Major published guidelines and research in the field acknowledged and critically appraised. All referencing appropriate for academic submissions.